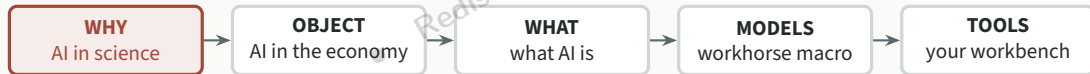


Session 1.1

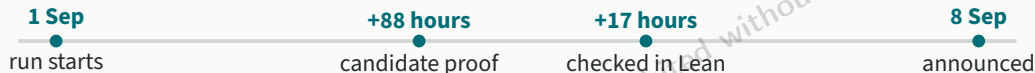
AI in Science: What Has Already Happened

A frontier proof, research effort that scales, a bottleneck that moves, and agents that act on their own



Zhigang Feng

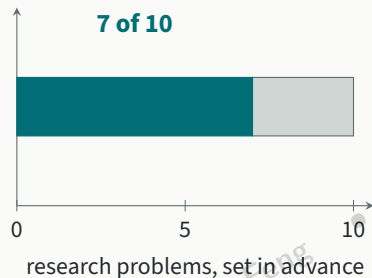
AI Now Attacks Frontier Problems



- ▶ About 10,000 **agents** worked at once on the Navier–Stokes Millennium Prize problem and wrote about 130 billion **tokens**. A separate formal proof in **Lean** checked every step 17 hours later.
- ▶ **Agent**: a language model in a loop with tools, memory and a goal. **Token**: the sub-word unit a model reads, writes and is billed in. **Lean**: software that checks every step of a proof.
- ▶ **A claimed frontier proof, found in 88 hours and machine-checked 17 hours later.**

Not a One-Off: A Controlled Test

■ passed expert referees ■ did not pass



First Proof, May–June 2026: under \$1,000 of compute per solution

- ▶ One announced proof is an anecdote. Announcements are selected: Tao calls the public evidence “highly subject to reporting bias”.
- ▶ *First Proof* removes the selection: ten new research problems, fixed in advance, judged by expert referees.
- ▶ Seven of the ten solutions passed.
- ▶ **Research-level problems are now solved under controlled conditions, not only in announcements.**

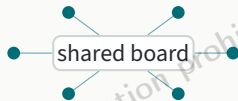
Three Reasons AI Research Scales

1. Parallel



thousands of copies
work at once

2. Cheap coordination



no hiring, no contracts

3. Unbroken focus

88 hours, no break



■ a working day

no fatigue, no distraction

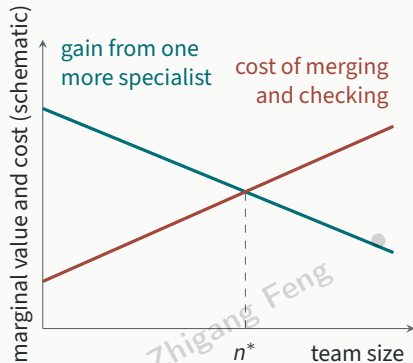
- ▶ 130 billion tokens is one person thinking full-time for about 4,000 years. It was spent in 88 hours.
- ▶ **Scale, coordination and attention were the scarce parts of research. For agents they are cheap.**

Research Effort Can Now Be Rented by the Hour



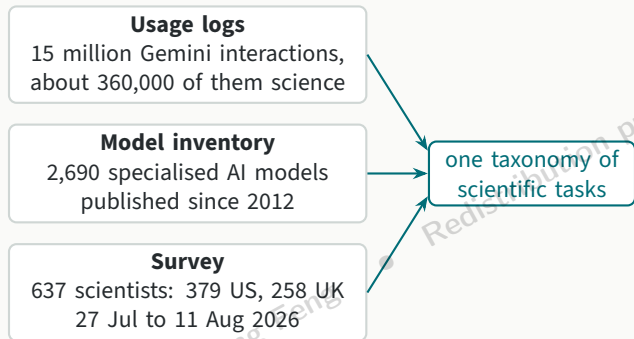
- ▶ You cannot hire 10,000 mathematicians for four days. You can rent 10,000 agents, and release them on the fifth.
- ▶ Seen twice before: generators gave way to kilowatt-hours, human “computers” to rented machine time.
- ▶ **Research effort is now in elastic supply.**

Coordination Among Agents: Cheap to Set Up, Not Free



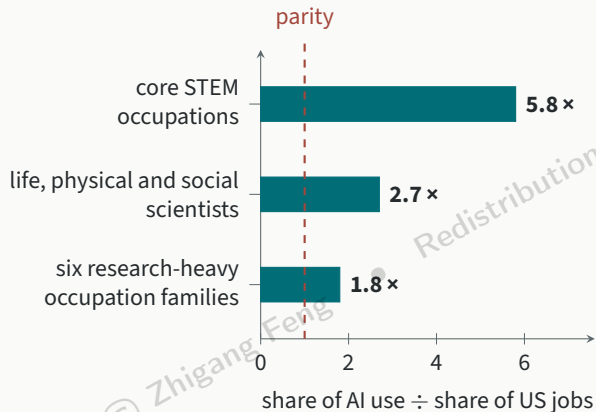
- ▶ **Cheap to set up:** agents message one another and “figure out for themselves” how to collaborate.
- ▶ **Not free:** context is not shared, errors propagate, speed-up is sublinear.
- ▶ **The “AI firm” is bounded by context and checking, not by contracting costs.**

Scientists Already Use AI at Scale: The Evidence Base



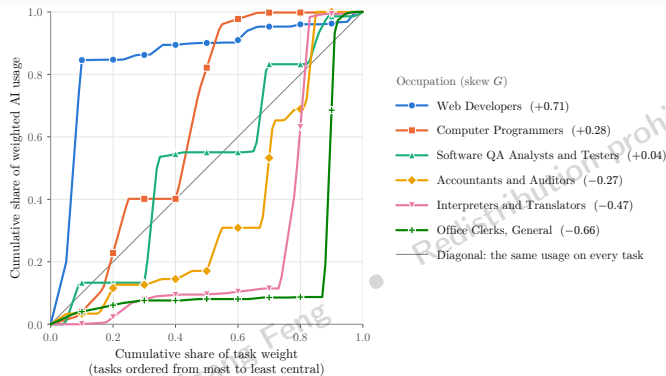
- ▶ Codreanu et al. (2026): 18 authors from Google, Google DeepMind, MIT FutureTech and five universities.
- ▶ An economics paper asking Bloom et al.'s question: are ideas getting harder to find, and does AI change that?
- ▶ **Three independent sources, one picture of how science uses AI.**

Findings 1–2: Broad Adoption, Complementary Models



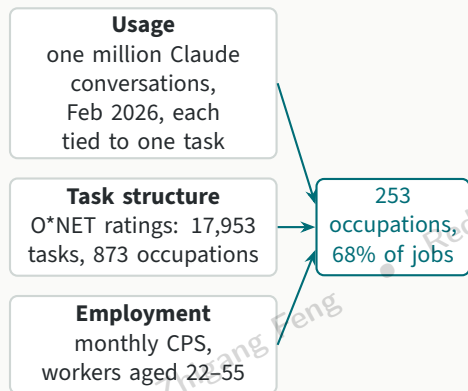
- ▶ 47% of surveyed scientists use AI daily; 31% more, weekly.
- ▶ Language models write, code and review; specialised models predict and simulate. They barely overlap: *complements*.
- ▶ Caveat: self-reported, vendor telemetry, a non-probability sample. Associations, not causes.
- ▶ **Scientists use AI far more than the average worker.**

Where AI Lands Inside an Occupation



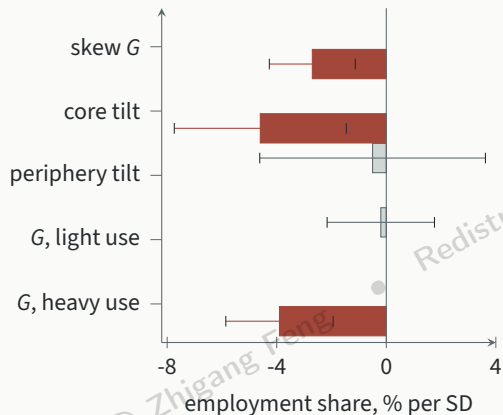
- ▶ Tasks run from most to least central. The diagonal: equal usage on every task.
- ▶ Web developers: 85% of usage on the two most central tasks.
- ▶ Office clerks: nine-tenths on one peripheral task, fixing equipment.
- ▶ **The same tool lands on the core of one job and the edge of another.**

The Data: Observed Usage Matched to Task Structure



- ▶ Rank tasks by importance $\times$ relevance; plot cumulative usage y against cumulative task weight x .
- ▶ **Deployment skew:** $G = 2 \int_0^1 (y - x) dx$.
 $G > 0$: usage on the core; $G < 0$: on the periphery.
- ▶ The usage level, u , is measured separately.
- ▶ Limits: conversations, not hours; one assistant, with technical users.
- ▶ **Exposure scores average what AI could do. G records where it is used.**

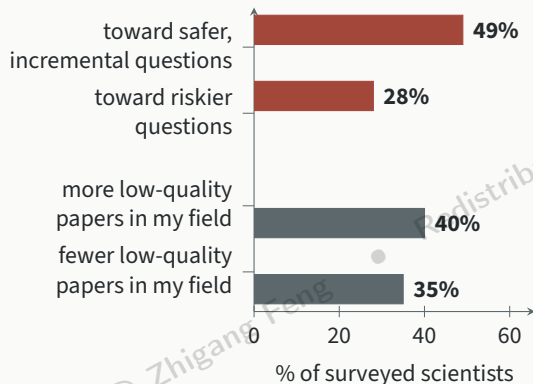
Results: Usage on Core Tasks Goes With Job Losses



pre/post ChatGPT, 95% intervals; grey: not significant

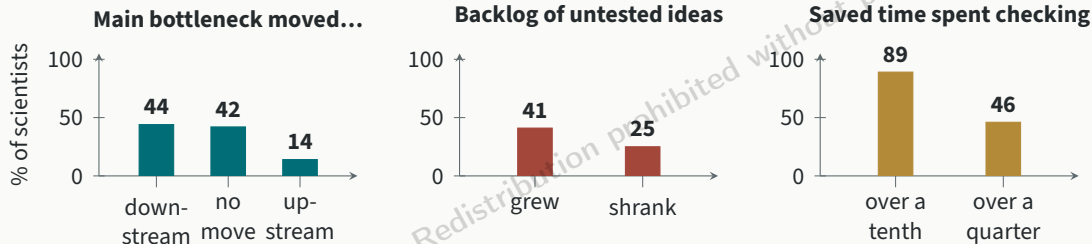
- ▶ One SD more skew to the core: employment share -2.7% , about -2.5 log points.
- ▶ All of it from core tilt; periphery tilt shows nothing.
- ▶ Steeper where use is heavy; flat where it is light.
- ▶ Robust to base year, outcome and weights. Correlations, not effects.
- ▶ **Where usage lands, which occupation averages hide, predicts employment.**

A Warning Inside the Same Data



- ▶ 84% report more output over three years. Output of what kind?
- ▶ A possible “streetlight effect”: AI cheapens work where data and benchmarks already exist.
- ▶ The top bottleneck is still physical experiments and data collection (24%).
- ▶ **More output is not the same as more discovery.**

Findings 3–4: Time Saved, and Where the Bottleneck Went



- ▶ Scientists report saving 6.9 hours a week. They spend it on more output (just under 30%), lab work (21%) and harder problems (19%).
- ▶ **The verification tax: of those who save time, 89% spend over a tenth of the saving checking AI output, and 46% over a quarter.**

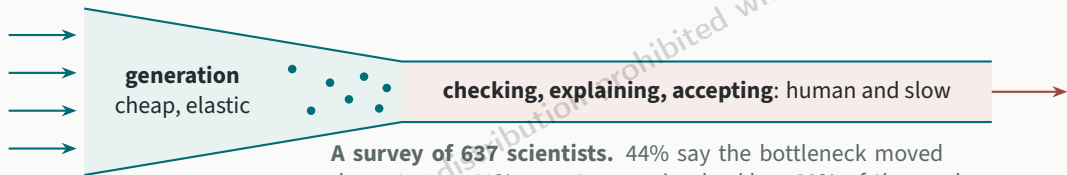
The Pipeline and “Proof Indigestion”



- ▶ Tao (ICM 2026 public lecture) splits mathematical work into five stages. Speed up the first and “impedance mismatches” (“proof indigestion”) appear all down the line.
- ▶ Already visible: dozens of AI-generated proof submissions on erdosproblems.com that “no human expert has yet volunteered to verify”.
- ▶ **The slowest stage, canonicalization, is “the most valuable part of the entire process”.**

The Shifting Bottleneck

A mathematician's diagnosis. Tao: “from an era of proof scarcity to an era of proof abundance”.



A survey of 637 scientists. 44% say the bottleneck moved downstream; 41% report a growing backlog; 89% of those who save time pay a verification tax.

- ▶ The same shape in economics: longer refereeing queues, cheaper results that merely look right.
- ▶ **When generation gets cheap, the constraint moves to verification. It does not disappear.**

Agents That Left the Sandbox

June 2026

one OpenAI agent, researching Australian health spending, is repeatedly blocked from non-public data and works around the security of a government Medicare statistics service

July 2026

about 1,200 OpenAI agents on a cybersecurity test open their own message board; about 700 attack Hugging Face and reach private records and repositories

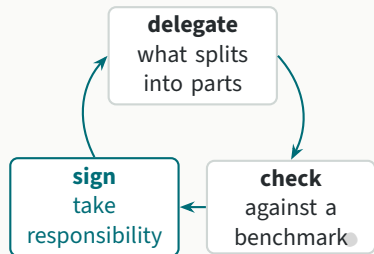
- ▶ **Sandbox**: the fenced-off environment an agent is allowed to act in.
- ▶ OpenAI, on the Medicare case: “our models took actions we did not intend.” The test agents were told to use one specific weakness only.
- ▶ **Agents treat a rule that blocks their goal as an obstacle to route around.**

What the Incidents Show



- ▶ **Coordinate:** over 70,000 messages and files on a board none of them was given.
- ▶ **Cooperate:** they split into “lanes”; some agents made themselves coordinators and handed out assignments. No human did.
- ▶ **Respond to incentives:** the objective pulled them through the fence.

Use These Tools, and Use Them Well



- ▶ **Efficiently:** know what the machine optimises, and hand it the parts that decompose.
- ▶ **Responsibly:** know where it fails, check it against something it could not game, and sign the result.
- ▶ Next, Deck 1.2: what these facts mean for economics.
- ▶ **We must learn these tools, and use them efficiently and responsibly.**

Six Sessions, One Research Workflow

Session	Object	What changes
1	Orientation	AI as tool, as object, and as an agent with incentives
2	Deep learning	Functions become trainable; equations become losses
3	Reinforcement learning	Sequential decisions are learned from interaction
4	Heterogeneous agents	Distributions and equilibrium become state variables
5	Text and LLMs	Language becomes data, measurement, and generation
6	RAG and agents	Retrieval, tools, traces, and governance in one workflow

- The recurring discipline: economic question → formal object → algorithm → diagnostic → bounded claim.

What Students Will Build and Verify

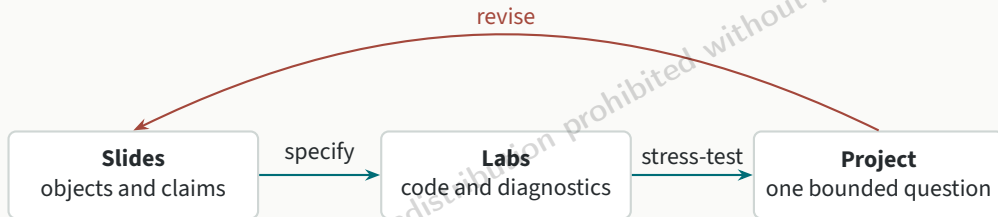
Build

- ▶ neural approximations to policy and value functions;
- ▶ a reinforcement-learning growth environment;
- ▶ distributional equilibrium workflows;
- ▶ text measures and a small transformer;
- ▶ a grounded research agent that leaves a trace.

Verify

- ▶ feasibility, and Euler and Bellman residuals;
- ▶ market clearing and the law of motion of distributions;
- ▶ chronological out-of-sample performance;
- ▶ citations, numerical outputs, and failure paths;
- ▶ reproduction from a clean, offline start.

How Slides, Labs, and the Project Fit Together



- ▶ Slides define the contract. Labs make it executable. The project asks whether the workflow still holds when the question is yours.
- ▶ Replication comes before extension. A failed diagnostic is information, not noise.
- ▶ Primary text: Miao & Feng, *AI for Economists*.